Luan Alfredo Lugarini

Computer Engineering Student

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Lattes CV

Fourth-semester Computer Engineering student at PUCPR seeking an engineering internship. I built and maintain web systems used daily by two family businesses (Next.js, TypeScript and Supabase). At university, projects in C, Python and robotics (Arduino and LEGO SPIKE). Advanced English (C1).

Education

2025 – 2029 **B.S. in Computer Engineering**, *Pontifical Catholic University of Paraná (PUCPR)*, Curitiba, Brazil, 4th semester, expected graduation 2029

2021 – 2023 **High School**, *Colégio Imaculada Conceição*

Experience

2026 – Present **Full-Stack Developer**, *Madeireira Lugarini (family business)*, Campo Magro, Brazil

- Built the quotes and orders app (PWA) the company uses daily: price calculation by lumber volume, installment payments, PDF generation and sending quotes via WhatsApp.
- Stack: Next.js 14, TypeScript, Tailwind, Supabase (PostgreSQL, auth and RLS), deployed on Vercel.
- Reliability and performance work: atomic payment recording in the database (RPC function) to prevent duplicate entries, and a 36 to 41% bundle reduction by dynamically importing the PDF library.
- Restructured the company website (madeiraslugarini.com.br) in HTML, CSS and JavaScript, including a new catalog page.

2026 – Present **Full-Stack Developer**, *Motel Grand Place (family business)*, Curitiba, Brazil

- Replaced the financial closing spreadsheet with a web system the owner uses on phone and desktop: daily entries in an editable grid, employee payments and monthly closing.
- Next.js 14 and Supabase, with database access policies (RLS) and import of the spreadsheet's historical data.

Academic Projects

2025 **MyArm Robotic Arm: Tower of Hanoi**, *Principles of Robotics, PUCPR*, C++ / Arduino · GitHub

- Assembled on my own a 4-degree-of-freedom robotic arm (base, two joints and gripper) driven by servo motors and an Arduino.
- Programmed the motion sequence that physically solves the Tower of Hanoi puzzle, with servo speed control for smooth movement.

2025 **Autonomous Maze-Solving Robot**, *Principles of Robotics, PUCPR*, Python / LEGO SPIKE · GitHub

- Robot that navigates a physical maze unassisted, using a distance sensor (walls), a color sensor (floor markers) and a gyroscope (90° and 180° turns with angle correction).
- Logic built as a state machine (outbound, return, finish) using asynchronous programming.

2026 **Conway's Game of Life**, *Imperative Programming, PUCPR*, C · GitHub

- Cellular automaton simulator with a dynamically allocated grid (`malloc`/`calloc`), initial state loaded from file, and dead borders.
- Double buffering with pointer swapping between generations, avoiding data copies and incorrect in-place updates.

2026 **ppm2ascii: Image to ASCII Art Converter**, *Imperative Programming, PUCPR*, C · GitHub

- Full parsing and validation of binary PPM (P6) images, supporting 8 and 16 bits per channel, with overflow checks before allocation.
- Per-pixel luminance calculation mapped onto a 57-character ramp.

2026 **15 Puzzle**, *Imperative Programming, PUCPR*, Python · GitHub

- 4x4 sliding puzzle in the terminal, with a shuffle that always produces a solvable board and an auto-solver that replays the move history as a stack.

Skills and Languages

Languages	TypeScript, JavaScript, C, Python, C++ (Arduino), SQL, HTML/CSS
Web	Next.js, React, Tailwind, Supabase (PostgreSQL), Vercel
Tools	Git, GitHub, VS Code, Arduino (servo motors), LEGO SPIKE (distance, color and gyro sensors)
Spoken	Portuguese (native); English, advanced (C1), certified by Wise Up